

GAZANIA MARINE JYRWA

Indian Institute of Technology, Jodhpur, India

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RESEARCH INTERESTS

Scientific Machine Learning; Neural Operators and Physics-Informed Architectures for Dynamical Systems; Foundation Models for Robotics and Control Systems

RESEARCH EXPERIENCE

Indian Institute of Science, Bangalore

November 2025 – March 2026

Research Associate, Advisor: Prof. Sashikumaar Ganesan

- Proposed and co-developed the Adaptive Wavelet-Fourier Neural Operator (AW-FNO), a hybrid neural operator architecture.
- Core contributor to SciREX (scientific computing & ML library): designed modular, scalable architecture and maintained repository for research and industrial use.
- Developed data-generation pipelines for Scientific Foundation Model, converting simulation outputs into structured mathematical/statistical representations and aligning them with LLM-generated physics-aware explanations

Project Assistant, Advisor: Prof. Sashikumaar Ganesan

August 2025 – November 2025

- Developed Geometry-Informed Neural Operators (GINOs) for an industry collaboration, enabling accurate magnetic field prediction in stator geometries

Indian Institute of Technology Jodhpur

August 2024 – May 2025

MSc Project, Advisor: Prof. Kirankumar R. Hiremath

- Conducted a comparative study of Finite Difference Methods (FDM) and Physics-Informed Neural Networks (PINNs) for solving the Helmholtz equation (elliptic PDE), evaluating accuracy, convergence, and computational efficiency through systematic numerical experiments.

WORK EXPERIENCE

ZenteiQ AiTech Innovations Pvt. Ltd.

April 2026 – Present

SciML Engineer

- Implementing Transformer-based PDE solvers, Transolver in JAX and exploring physics-attention mechanisms for learning over spatial point representations, while contributing to production-grade SciML models for industrial applications.

CONFERENCE CONTRIBUTIONS

AW-FNO: Adaptive Wavelet-Fourier Neural Operator for Fluid Flow Super-Resolution

Abstract accepted at the 13th International Conference on Computational Fluid Dynamics (ICCFD13), 2026, full paper in preparation

EDUCATION

Indian Institute of Technology, Jodhpur

August 2023 - May 2025

Master of Science (M.Sc) in Mathematics

CGPA: 6.29/10

North Eastern Hills University, Shillong

August 2020 - July 2023

Bachelor of Science (B.Sc) in Mathematics

Percentage: 65.5

TECHNICAL SKILLS

- **Languages:** Python, C++
- **Libraries & Frameworks:** PyTorch, JAX, TensorFlow, NumPy, Pandas, Matplotlib, SciPy
- **Deep Learning & Computer Vision:** CNNs, Vision Transformers, image analysis, attention mechanisms, representation learning
- **Scientific Computing & SciML:** Neural Operators, PINNs, FDM, GPU computing, CUDA, Gmsh
- **Tools & Workflow:** Git, GitHub, Github Actions, Docker, Linux, ClearML, L^AT_EX, VSCode, Jupyter

HIGHLIGHTED PROJECTS

Transolver: 2D Turbulence & Aerodynamic RANS Surrogates

Github

Built a physics-aware computational surrogate framework benchmarking Transolver against FNO on 2D Navier–Stokes turbulence and unstructured-grid RANS aerodynamics

- Transolver achieved 0.40 % relative L error at 183k parameters vs FNO’s 0.46 % error at 1.42M parameters a 1.5× accuracy gain with 10× parameter efficiency

Vision Transformer for Image Representation Learning

Github

Implemented a Vision Transformer from scratch for CIFAR-10 image classification using PyTorch.

- Built the complete ViT pipeline including patch embeddings, positional encodings, multi-head self-attention, transformer encoder blocks, and end-to-end training/evaluation.

PINSTT: Physics-Informed Neural Spatiotemporal Tubes

Github

Reproduced paper arXiv:2512.08248 from scratch: learning time-varying safety tubes for robot navigation using PINNs with formal verification guarantees

- Implemented a three-stage pipeline: PINN training, Lipschitz-based formal verification, and a log-barrier online controller for Temporal Reach-Avoid-Stay (T-RAS) tasks.

Helmholtz Equation Solver: PINNs vs FDM Github

Github

Implemented and compared Physics-Informed Neural Networks (PINNs) and Finite Difference Methods (FDM) for solving the Helmholtz equation

- Analyzed error, convergence, and computational efficiency against analytical solutions.

Image Generation with DDPM

Github

Implemented a Denoising Diffusion Probabilistic Models-based image generation pipeline on CIFAR-10

- Developed the complete diffusion workflow including forward noising, reverse denoising, U-Net based noise prediction, training/sampling scripts, checkpoints, visualization, and Docker-based reproducible setup

RELEVANT COURSEWORK

Mathematics: Linear Algebra*, Optimization*, Probability & Statistics*, Vector Calculus, ODEs*, PDEs*, Numerical Methods, Dynamical Systems*

Computational Science & Programming: Digital Image Analysis*, Machine Learning and Pattern Recognition*, Programming Techniques*, Data structures & Algorithms*

Online Courses: Machine Learning Specialization (Coursera), Neural Networks from Scratch (Udemy), Python for Data Science and Data Analysis (Udemy), Practical Deep Learning (Udemy)

(* denotes postgraduate-level subjects taken as major coursework)

EXTRA-CURRICULARS

- Captain, Women’s Football Team, IIT Jodhpur — Runners-up, Varchas (2024)
- Active athlete with multiple medals in collegiate and school sports and athletics.